



Lesson 5.6 — Solving Radical Equations

Big idea: To solve a radical equation, isolate the radical and raise both sides to the right power. ALWAYS check answers — squaring can introduce extraneous solutions.

Quick review

Step 1: isolate the radical on one side.

Step 2: raise both sides to the index power (square for $\sqrt{\quad}$, cube for $\sqrt[3]{\quad}$).

Step 3: solve the resulting equation.

Step 4: CHECK every answer in the original equation.

Why check: squaring is not reversible — it can create solutions that don't satisfy the original.

Worked example

Worked example. Solve $\sqrt{2x + 5} = x - 1$.

Square: $2x + 5 = (x - 1)^2 = x^2 - 2x + 1$. Rearrange: $x^2 - 4x - 4 = 0$. $x = (4 \pm \sqrt{32})/2 = 2 \pm 2\sqrt{2}$. Check: $2 + 2\sqrt{2} \approx 4.83$ ✓; $2 - 2\sqrt{2} \approx -0.83$, but RHS = $x - 1 \approx -1.83$ negative while LHS is non-negative ✗ extraneous.
Solution: $x = 2 + 2\sqrt{2}$.

Warm-ups

1. Solve $\sqrt{x} = 4$.
2. Solve $\sqrt[3]{x} = -2$.
3. Solve $\sqrt{x - 1} = 3$.

Practice

1. Solve $\sqrt{3x + 1} = 5$.
2. Solve $\sqrt{x} = x - 6$.
3. Solve $\sqrt[3]{2x + 5} = 3$.

Challenge

1. Solve $\sqrt{x + 6} = \sqrt{x - 2} + 2$.
2. Solve $x - \sqrt{x - 2} = 4$.



Answer key — Lesson 5.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 16$.
2. $x = -8$.
3. $x - 1 = 9 \Rightarrow x = 10$. Check: $\sqrt{9} = 3$ ✓.

Practice

1. Square: $3x + 1 = 25 \Rightarrow x = 8$. Check: $\sqrt{25} = 5$ ✓.
2. Square: $x = (x - 6)^2 = x^2 - 12x + 36 \Rightarrow x^2 - 13x + 36 = 0 \Rightarrow (x - 4)(x - 9) = 0$. Check $x = 4$: $\sqrt{4} = 2$, but $\text{RHS} = 4 - 6 = -2$ ✗. Check $x = 9$: $\sqrt{9} = 3 = 9 - 6$ ✓. Solution: $x = 9$.
3. Cube: $2x + 5 = 27 \Rightarrow x = 11$.

Challenge

1. Square: $x + 6 = (x - 2) + 4\sqrt{(x - 2) + 4} = x + 2 + 4\sqrt{(x - 2)} \Rightarrow 4 = 4\sqrt{(x - 2)} \Rightarrow \sqrt{(x - 2)} = 1 \Rightarrow x = 3$. Check: $\sqrt{9} = \sqrt{1 + 2} = 3$ ✓.
2. Isolate: $\sqrt{(x - 2)} = x - 4$. Square: $x - 2 = (x - 4)^2 = x^2 - 8x + 16 \Rightarrow x^2 - 9x + 18 = 0 \Rightarrow (x - 3)(x - 6) = 0$. Check $x = 3$: $3 - 1 = 2 \neq 4$ ✗. Check $x = 6$: $6 - 2 = 4$ ✓. Solution: $x = 6$.